

Project6

对比 2 个终端

```
star@star-A:~/Desktop/RoboticBasic/Lecture6$ python3 test inverse kinematics.py
[INFO] [1784888158.814661338] [manipulator]: Press Ctrl + C to terminate
[INFO] [1784888158.817786241] [manipulator]: Test case (target position): [0.15, 0.0, 0.25]
[INFO] [1784888158.817917079] [manipulator]: Method          waist  shoulder  elbow
[INFO] [1784888158.818023601] [manipulator]: -----
[INFO] [1784888158.818133807] [manipulator]: Analytic          0.0000   -0.3187   0.3995
[INFO] [1784888158.818241121] [manipulator]: Numerical         0.0000    1.2730  -2.8976
[INFO] [1784888158.818348701] [manipulator]: Publishing: Analytic
^CShutting down the manipulator node...
star@star-A:~/Desktop/RoboticBasic/Lecture6$ python3 test inverse kinematics.py
[INFO] [1784888198.900790596] [manipulator]: Press Ctrl + C to terminate
[INFO] [1784888198.903701855] [manipulator]: Test case (target position): [0.15, 0.0, 0.25]
[INFO] [1784888198.903848735] [manipulator]: Method          waist  shoulder  elbow
[INFO] [1784888198.903955513] [manipulator]: -----
[INFO] [1784888198.904067434] [manipulator]: Analytic          0.0000   -0.3187   0.3995
[INFO] [1784888198.904176995] [manipulator]: Numerical         0.0000    1.2730  -2.8976
[INFO] [1784888198.904285173] [manipulator]: Publishing: Numerical
```

通过 publish_joint_state 传递参数，得出的位置是一样的

```
star@star-A:~/Desktop/RoboticBasic/Lecture6$ ros2 run tf2_ros tf2_echo rx150/base_link rx150/wrist_link
[INFO] [1784887930.348651655] [tf2_echo]: Waiting for transform rx150/base_link -> rx150/wrist_link: Invalid frame ID
"rx150/base_link" passed to canTransform argument target_frame - frame does not exist
At time 1784887931.332956357
- Translation: [0.150, 0.000, 0.251]
- Rotation: in Quaternion (xyzw) [0.000, 0.040, 0.000, 0.999]
- Rotation: in RPY (radian) [0.000, 0.081, 0.000]
- Rotation: in RPY (degree) [0.000, 4.632, 0.000]
- Matrix:
  0.997  0.000  0.081  0.150
  0.000  1.000  0.000  0.000
 -0.081  0.000  0.997  0.251
  0.000  0.000  0.000  1.000
At time 1784887932.332804024
```